Sao Cheng Tai

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MEng Computing (AI & ML) student at Imperial College London focused on ML research and data science. Fine-tuned MedSAM2 with LoRA (0.969 Dice, >90% lower training cost), developing a PINN that reconstructs hidden wall shapes from pressure data in supersonic flow, and London AI x Science Hackathon finalist (top 8 of 50+). Eager to apply rigorous experimentation to AI for science, data science and computer vision.

EDUCATION

Imperial College London *2026 – 2030*
MEng Computing (Artificial Intelligence and Machine Learning)

Yuet Wah College, Macau *2020 – 2026*
- International A Levels: 4A* (Mathematics, Further Mathematics, Physics, Chemistry)

RESEARCH EXPERIENCE

Physics Informed Neural Network (PINN) Shape Inversion in Supersonic Flow – *Student Researcher 2026 - now*
- Developing a PINN to reconstruct hidden wall shapes from wall pressure and shear-stress data.

Fast & Light: Low-Rank Adaptation for Medical Imaging – *Student Researcher (team of six)* *2025 – 2026*
- Fine-tuned MedSAM2, a SAM2-based medical segmentation model, with LoRA on 1.3% of its parameters, reaching 0.969 Dice in 72 minutes and cutting training time and cost by over 90%.
- Designed a dual-prompt fusion strategy using automated wide and central bounding boxes to handle anatomical challenges such as cardiac obscuration in chest X-rays.

HACKATHONS & SOFTWARE PROJECTS

London AI x Science Hackathon – *Finalist, Polaron Materials Manufacturing Track (team of five)* *2026*
- Built a full-stack AI application that classifies SEM images and analyses classification logic.
- Ranked in the top 8 of 50+ submissions (top 25 of 1,200+ applicants) against teams made up mainly of PhD and master's students, researchers, engineers and scientists.
- Built a multi-agent explainer with Corrective RAG (Claude API): analyst, research-librarian, writer and reviewer agents explain vision model predictions, cite research, and auto-check cited information; the reviewer flagged 32% of draft claims for revision before output during testing.
- Designed a vision pipeline: distilled a LightGBM model into a U-Net that segments material phases (0.3 s/image); combined with DINOv2 texture features to cross-check the main batch classifier.

Post-Disaster Debris Detection System – *Project Manager & Software Lead (team of six)* *2026*
- Built an AI + GIS debris-response system for post-disaster Macau: DEM watershed analysis flags likely accumulation zones, and YOLOv12 detections feed a bilingual live heatmap dashboard.
- Trained a YOLOv12 detector (10,381 images, 11 classes, 86.8% mAP@50, 89.5% precision).
- Built a FastAPI inference service and a real-time camera pipeline with GPS-tagged logging and fault-tolerant inference with timeouts and exponential-backoff retries.

LLM Authorship Attribution | Kaggle "Name That LLM" – *Individual project* *2026*
- Built an NLP pipeline to distinguish 18 LLMs using 2,699 passages across 150 prompts.
- Combined TF-IDF, 176 stylometric features, and 4,108 frozen ELECTRA-derived features.
- Ensembled SVM, CatBoost, and logistic regression, achieving 45.19% local validation balanced accuracy, 8.89 percentage points above baseline.
- Evaluated 12 classifier configurations with prompt-grouped five-fold validation; built reproducible caching, automated tests, and offline Kaggle notebooks.

HONOURS & AWARDS

- **First Runner-Up ×2**, Youth Science & Technology Village (Engineering; Biology & Chemistry) *2026*
- **Macau PhD Think Tank High-Tech Scholarship** *2026*
- **Gold Award ×2**, National Youth AI Innovation Challenge (Inspection Robot; Smart Farm) *2025, 2023*
- **First Prize ×2**, Macau AI Challenge (1st and 3rd editions) *2025, 2023*

TECHNICAL SKILLS

Languages & Software Engineering: Python, C++, JavaScript, FastAPI, REST APIs, React, Git, pytest, Arduino
ML & Data Science: PyTorch, sklearn, LightGBM, XGBoost, NumPy, SciPy, pandas, Matplotlib, SQL (SQLite)
LLMs & Agents: Claude & OpenAI API, multi-agent system, LangChain, RAG (e.g. Corrective), LoRA fine-tuning
Computer Vision: OpenCV, Ultralytics YOLO, Roboflow, U-Net, DINOv2, MedSAM2, knowledge distillation